# Deliverable 4: System Requirements

## System Requirements Document

System Requirements (1).docx

The System Requirements Document (SRD) translates the selected system architecture and trade study decisions into solution-dependent, verifiable system requirements.

### SYS.1.1 Dual-Stream Capture

The system shall capture synchronized eye-tracking and scene video streams at a minimum sampling rate of 200 Hz.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.2 Data Synchronization

The system shall maintain synchronization between eye and scene video streams with a maximum timestamp deviation of 10 milliseconds.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.3 Operational Gaze Accuracy

The system shall compute gaze coordinates from captured data with an angular accuracy of ±2 degrees.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.4 Data Upload

The system shall upload inspection session data to the cloud within 10 seconds under standard network conditions.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.5 Inspection Analytics

The system shall generate gaze heatmaps and scanpath visualizations aligned with predefined SOP inspection zones.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.6 Battery Performance

The system shall operate continuously for at least 6 hours on a single battery charge.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.7 Calibration Accuracy

The system shall provide a calibration process that achieves a gaze estimation error of ±1° under controlled conditions.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.8 Inspection Coverage

The system shall calculate the percentage of inspection zones visually inspected during a session.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.9 Hands-Free Operation

The system shall operate without requiring manual interaction during inspection.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.10 Inspection Report

The system shall generate an inspection report containing heatmaps, scanpaths, and inspection coverage metrics.

Rationale:

Quality Score: 0%

Labels: *None to display.*

### SYS.1.11 Network Reliability

The system shall successfully upload inspection data with a success rate of ≥95% under standard network conditions.

Rationale:

Quality Score: 0%

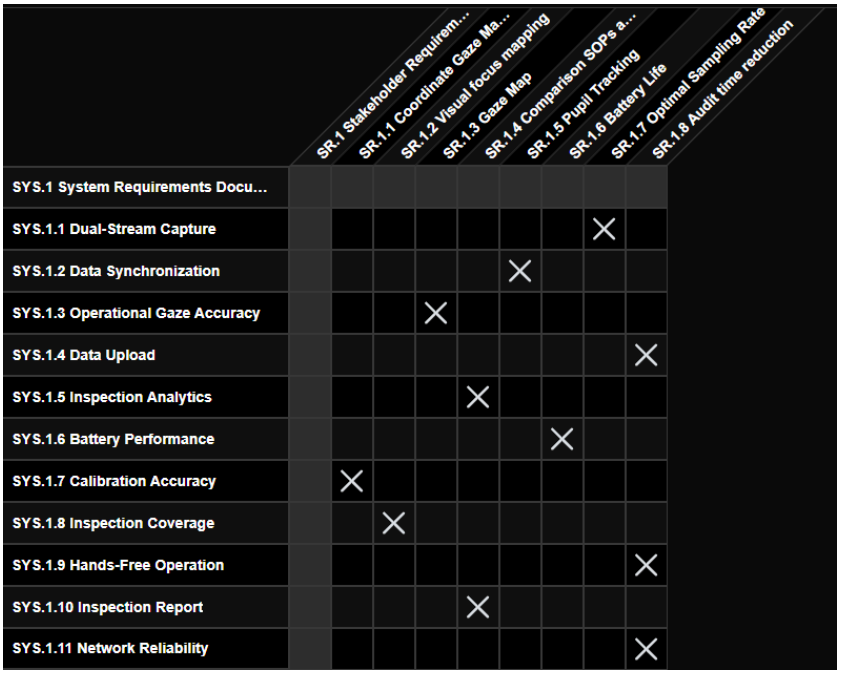
Labels: *None to display.*

# Traceability matrix:

Matrix_Report (1).xlsx

The above file only showed empty columns, therefore using the csv file generated from innoslate.

System_Requirement_Traceability.csv




| | SR.1 Stakeholder Requirements | SR.1.1 Coordinate Gaze Mapping | SR.1.2 Visual focus mapping | SR.1.3 Gaze Map | SR.1.4 Comparison SOPs available | SR.1.5 Pupil Tracking | SR.1.6 Ba Life |
|---|---|---|---|---|---|---|---|
| SYS.1 System Requirements Document | | | | | | | |
| SYS.1.1 Dual-Stream Capture | | | | | | | |
| SYS.1.2 Data Synchronization | | | | | | X | |
| SYS.1.3 Operational Gaze Accuracy | | | | X | | | |
| SYS.1.4 Data Upload | | | | | | | |
| SYS.1.5 Inspection Analytics | | | | | X | | |
| SYS.1.6 Battery Performance | | | | | | | X |
| SYS.1.7 Calibration Accuracy | | X | | | | | |
| SYS.1.8 Inspection Coverage | | | X | | | | |
| SYS.1.9 Hands-Free Operation | | | | | | | |
| SYS.1.10 Inspection Report | | | | | X | | |
| SYS.1.11 Network Reliability | | | | | | | |